

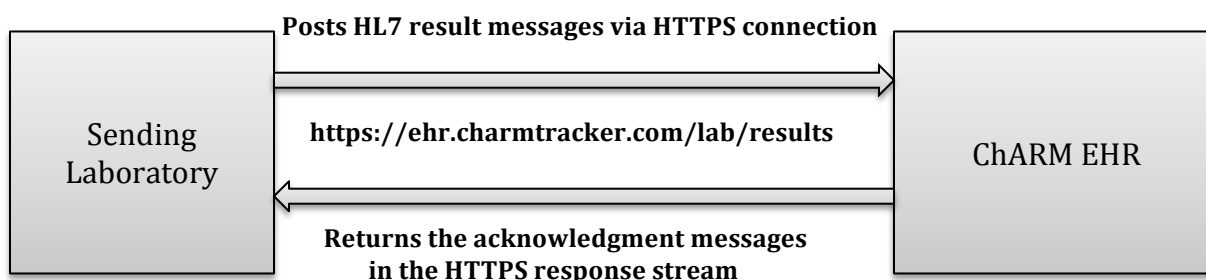
ChARM EHR Lab Result Interface Connectivity

Introduction

This document describes the connectivity details for ChARM EHR result interface. Labs can use this interface to submit Lab result messages in HL7 format to ChARM EHR. This document does not deal with the HL7 message details. Refer to “ChARM EHR UNI-DIRECTIONAL LAB INTERFACE IMPLEMENTATION GUIDE” for HL7 message specification.

Connectivity

ChARM EHR uses HTTPS protocol to transport data to and from Labs. HTTPS is a secure(SSL/TLS is layered above HTTP), reliable and widely used protocol for the exchange of data over TCP/IP. The below flow diagram represents the connectivity of labs and ChARM.



Header Authentication

ChARM EHR requires that all data to be sent using Basic Authentication. The field “Authorization” in the HTTP header should have the username and password separated by colon(“:”), which is encoded using base64 encoding.

- Concatenate the username and password, separating them with a colon: “username:password”
- Convert that concatenation into a UTF8 encoded byte array
- Convert that array into a base64 encoded string.
- Specify the authentication type as 'Basic' by adding the word 'Basic' to it.
- Add that string to the Authorization header. It will look like:
Authorization: Basic base64String
- Send the request.

Example

Let us consider,

username = "adminuser" (and) password = "adminpass"

1. Concatenate username and password with colon as separator.
(adminuser:adminpass)
2. Convert that concatenation into a UTF8 encoded byte array. Encode the UTF8 byte array using base64 encoding system.
(YWRtaW51c2VyOmFkbWlucGFzcw==)
3. Append the encoded string to "Basic" with an empty space in between. Set the result string to Authorization header.
(Authorization: Basic YWRtaW51c2VyOmFkbWlucGFzcw==)
4. Send the request.

Sample Post:

URL - <https://ehr.charmtracker.com/lab/results>

Authorization: Basic YWRtaW51c2VyOmFkbWlucGFzcw==

Content-Type: text/plain

Host: xxx.xx.x.xxx

Content-Length: 1567

```
MSH|^~\&|Lab||ChARM_EHR|FAC123456789|20140623065300||ORU^R01|146c460f15
|P|2.3.1
PID|1||PAT0001||LastName^FirstName||19750625|M||15293 NW CASEY
DR^PORTLAND^OR^97229||(503)645-7572^^^^503^6457572
ORC|RE||LK765435|||||201406230653||Physician
NPI^LastName^FirstName^^^^^NPI
OBR|1|LK765435|LK765435|82627^DHEA-
SO4^L||201406161720|||||Physician
NPI^LastName^FirstName^^^^^NPI|||^^|201406230653||F
NTE|1|L|Patient Not Fasting
OBX|1|NM|82627^DHEA-SO4^L|25|ug/dL|37-224|L||F||201406161720|LF^Lab
Facility^City^State^Zip
ORC|RE||LK765435|||||201406230653||Physician
NPI^LastName^FirstName^^^^^NPI
OBR|2|LK765435|LK765435|82728^Ferritin^L||201406161720|||||Physici
an NPI^LastName^FirstName^^^^^NPI|||^^|201406230653||F
NTE|1|L|Patient Not Fasting
OBX|1|NM|82728^Ferritin^L|27|ng/mL|6-170||F||201406161720|LF^Lab
Facility^City^State^Zip
ORC|RE||LK765435|||||201406230653||Physician
NPI^LastName^FirstName^^^^^NPI
OBR|3|LK765435|LK765435|84439^Free
T4^L||201406161720|||||Physician
NPI^LastName^FirstName^^^^^NPI|||^^|201406230653||F
NTE|1|L|Patient Not Fasting
OBX|1|NM|84439^T4, Free^L|1.0|ng/dL|0.7-1.5||F||201406161720|LF^Lab
Facility^City^State^Zip
```

Domain and IP requirements

- The Fully Qualified Domain Name(FQDN) address will resolve to a public IP address.
- You must use the ChARM FQDN with public Domain Name System (DNS) lookups to support the dynamic resolution of the ChARM FQDN. Do not hard-code IP addresses in hosts files or within application software.
- ChARM has different environments for Testing and Production. Testing and Production setups have different user credentials.

Test and Production Environment Details

	Test	Production
URL	https://stage106.charmtracker.com/lab/results	https://ehr.charmtracker.com/lab/results
Credentials	Will be provided on request	Will be provided on request

Communication Errors and Data Errors

The errors could be in communication and the other in the data transfer. They are categorized as communication failure and HL7 message format failure.

- When a request fails, appropriate HTTP status code will be sent in the response. ChARM uses standard HTTP status codes.
- If the request is successful, HTTP OK(200) status will be send with HL7 ACK message in the response stream.
- Refer to HTTP error codes for more information of these codes.

Java code sample

Refer `PostHL7LabResultsToChARMEHR.java` for sample code.